LECTURE # 21

ESD for Sustainable Healthcare

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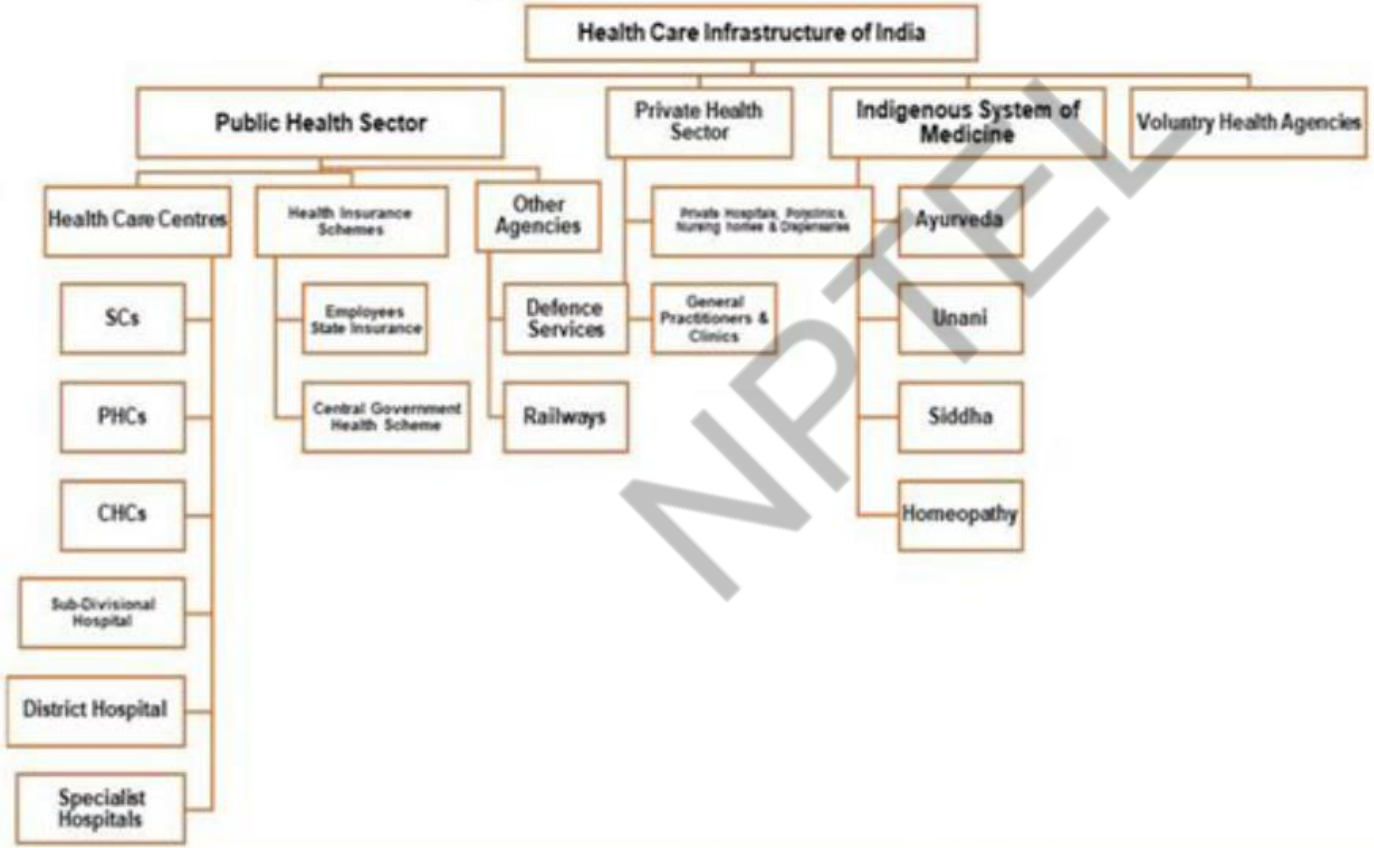
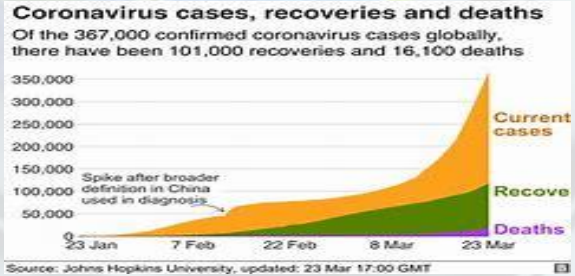

Health Care Infrastructure of India
Public Health Sector
Private Health Sector
Indigenous System of Medicine
Voluntry Health Agencies
Health Care Centres
Health Insurance Schemes
Other Agencies
Private Hospitals, Polyclinics, Nursing homes & Dispensaries
Ayurveda
SCs
Employees State Insurance
Defence Services
General Practitioners & Clinics
Unani
PHCs
Central Government Health Scheme
Railways
Siddha
CHCs
Homeopathy
Sub-Divisional Hospital
District Hospital
Specialist Hospitals

Problems of Healthcare Infrastructure in India

1. Shortage of Staff and Equipment
2. Healthcare infrastructure is heavily skewed in favour of urban areas
3. Inter-state and Intra-state variations
4. No proper regulatory mechanism and monitoring (Vij, 2019)

- A study by (Garg, et al., 2020) states that many lessons need to be learnt especially for developing economies like India where public healthcare system is grossly inadequate to take care of health needs of citizens post pandemic.
- This study utilized World Health Organization's framework of six health system building blocks to study the lessons learnt and actionable points in the post pandemic period.
- 1. Leadership/Governance
- 2. Health information System
- 3. Service Delivery
- 4. Financing
- 5. Health Workforce
- 6. Access to essential medicines



- A study which highlights disparities in health care infrastructure available both in rural and urban India by providing actual status of their availability in the country and need for their reallocation recommends
- strengthening of primary health centers
- increased expenditure on health by the government ensuring a perfect combination of physical and human infrastructure

➢**training and skill upgradation of local health** care workers, opening of medical colleges in rural areas, proper regulation and monitoring

➢ **stringent laws and punishment** for unethical practices in healthcare

➢ **use of technology like telemedicine** and **Mobile hospital in rural, hilly and tribal areas for** minimizing cost and **active** involvement of the **community in health care programs in to** achieve total health care of 1.3 billion people of the country (Vij, 2019).

- Participation in **Global Health Security Alliance** has to be stepped up with involvement in joint external evaluation and development of epidemiological core capacities
- **National Health Security Action Plan needs to drafted and available for health**

emergencies, **Ayushman Bharat scheme** (Health and Wellness Centers and National Health Protection Scheme) should incorporate elements to address surge capacity at the time of health emergencies and measures to deliver care at the time of pandemic

- **Technology through telemedicine, m-health, and digital platforms or apps should contribute to trainings, supervision, and facilitation of healthcare delivery at remote locations, open data sharing policies should be developed for the practice of evidence-based public health, public healthcare system and health manpower trained in epidemiology should be given a boost to have system readiness to respond in case of future pandemics.**

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- **Sustainability of a health and care system depends on seven factors**
- ***Internal factors***
- Efficiency and effectiveness of health and care provision
- Availability of well trained health and care workers
- Costs and economic benefits
- ***External factors***
- Health and resilience of the population
- Contribution of caregivers and informal networks of care
- Integration of policy and practice with other sectors and building healthy and health creating communities
- ***Overall***
- Public and political acceptability and support

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Value in healthcare

- Value in healthcare is simply a way of expressing **how well a healthcare system meets its objectives.**
- It can be defined as the **outcomes of a process, relative to the costs**.
- An **ecologically sustainable approach** broadens the definition of value.
- Costs include **environmental and social impacts**, not simply economic costs, giving what is referred to as a triple bottom line.
- Outcomes refers to both **individual patients** and **population outcomes**.
- Sustainable value in healthcare can be expressed as given below:

$$\text{Value} = \frac{\text{Outcomes for patients and populations}}{\text{Environmental + social + financial impacts}}$$

(the 'triple bottom line')

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What is Health Grid?

- A health grid has been defined as an environment in which data of medical interest can be stored and made easily available to different actors in healthcare systems such as physicians, healthcare centers, patients and citizens.
- Health grids focus equally on the sharing of data (and the associated issues of privacy and ethics) and on distributed health analysis across the biomedical spectrum from public health to patient care and from tissue/organ data to cellular and genomic information.
- For individualized healthcare, health grids are envisaged to facilitate access to biomedical information and ultimately knowledge, no matter where the requester of that information may reside or where the relevant data is stored: biomedical information on demand.
- Much research activity continues in the field of health grids (Manset, et al., 2009).

- https://www.bing.com/videos/search?q=ESD+for+sustainable+healthcare&ru=%2fvideos%2fsearch%3fq%3dESD%2bfor%2bsustainable%2bhealthcare%26go%3dSearch%26qs%3dds%26form%3dQBVDMH&view=detail&mid=1A14AC33611B12B8EBEA1A14AC33611B12B8EBEA&&FORM=VDRVSR

LECTURE # 22

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Envisioned Architecture of Global Public Health Grid (McNabb, 2010)

National Digital Health Mission (NHP 2020)

- **National Digital Health Mission will help create an integrated nationwide database of health service providers, patient records over care continuum and patient consent management using open API framework.**
- Technology thus aims to ensure proper **integration of available data** from all sources and make it available for **analytics**, and these dramatic changes in **healthcare data processing**, supported by **cloud-based technology ecosystems** and **healthcare innovations**, will provide newer understanding for **health care strategies in near future** and overall well being for entire India in times to come.
- **MISSION: aims to create a management mechanism to process digital health data and facilitate its seamless exchange; develop registries of public and private facilities, health service providers, laboratories and pharmacies; and to support clinical decision-making as well as offer services like telemedicine.**
- **Evidence based, transparent and efficient.**

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CURRENT SCENARIO

- India witnessed a shortage of hospital beds and oxygen supplies in both urban and rural settings during the COVID 19 pandemic.
- ● In a study by Gautam et al (2021) which focused to critically examine the existing healthcare facilities and the GOI's response to combat the COVID-19 apropos the NHP 2017 suggests policy options that can be adopted to prevent the further expansion of the pandemic and prepare the country for future health emergency-like situations. The results of this study showed that notwithstanding the several combatting steps on a war-footing level, COVID-19 has placed an extra burden over the already overstretched healthcare infrastructure.

IMPLICATIONS

- Government has implemented Anganwadi Services, Pradhan Mantri Matru Vandana Yojana and Scheme for Adolescent Girls under the Umbrella
- Integrated Child Development Services Scheme (ICDS) as targeted interventions to address the problem of malnutrition in the country.
- ● The POSHAN Abhiyan , launched in 2018, is India's flagship programme to improve nutritional outcomes for children, pregnant women and lactating mothers.
- ● A better and effectively planned coordination among ministries and organizations involved in the area is significantly very important.
- ● Realistic database of population suffering from nutritional deficiency, lacking in access to health infrastructure and other health related services need to be formulated, taking into

- Address all the dimensions of health and have professionals easily accessible to people
- ● Investments in health sector and social welfare schemes need to be increased.
- And centralized system needs to be created to keep a watch on distribution of the benefits of these schemes to those who are in need.
- ● More active and inclusive engagement at grassroots levels, such as those of Panchayats is essential in order to ensure effective implementation of health and welfare schemes.
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- https://www.bing.com/videos/search?q=sustainable+healthcare&ru=%2fvideos%2fsearch%3fq%3dsustainable%2bhealthcare%26FORM%3dHDRSC3&view=detail&mid=562F849B5E38FF7895E3562F849B5E38FF7895E3&rvsmid=6CF5E74772FD74AF0E0B6CF5E74772FD74AF0E0B&FORM=VDRVRV

Overall help of community people should be sought for providing health care to all because no programme can be successful without active involvement of community

Sustainable Quality Improvement (Sus-QI)

- Quality improvement (QI) is the ongoing, systematic effort to improve patient outcomes and system performance.
- Sustainable Quality Improvement (Sus-QI) is a practical framework that extends the scope of QI to include:
- • Ecological sustainability
- • Adding social value
- • A long-term perspective
- Sus-QI has a 4 stages approach that can be added to any /existing healthcare framework.

1• Set goals (e.g. minimizing waste in operating theatres, instituting waste segregation streams and recycling)

2• Study the system

3• Design the improvement effort

4• Measure impacts

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- The key difference with **Sus-QI** is in nature of the improvement efforts and the impacts measured.
- Improvements focus more on patient empowerment and self-care, prevention of disease, lean service delivery and low carbon alternatives.
- Impacts would include not only clinical impacts and financial costs, but also social impacts and environmental impacts.
- Ideally, when considering the environmental impacts of a system, one would calculate total ecological footprint, rather than simply estimating greenhouse gas emissions.
- However, we commonly see only carbon footprint reported, as this is much more easily quantifiable and reflects the current focus on the climate crisis.

- References:

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- **Efficiency and effectiveness of health and care provision**-Community and home based models of care already exist and can be enhanced with technology, new scientific discovery, and the engagement of patients, families, and communities
- **Availability of well trained health and care workers**-This requires vision, leadership, and a determined programme of training and development
- **Costs and economic benefits**-Recent thinking in health has emphasized the importance of measuring the value of procedures and services and of doing, in effect, cost-benefit analyses that compare outcomes with inputs. This approach needs to be extended to the whole system
- **Health and resilience of the population**-Health status is markedly different around the country, largely related to economic and social factors. Children's health and wellbeing in particular have long term effects on the health and care system and its sustainability.
- **Contribution of private caregivers and informal networks** of NGO-care agencies.

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LECTURE # 23

ESD for Sustainable Healthcare

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- **Integration of policy and practice with other sectors** and building healthy and health creating communities-Employers, educators, designers, planners, and others also determine health status and have responsibilities for protecting and promoting health.
- **Public and political acceptability and support**-the economic benefits derived from the health and care system and the potential for multi-sectoral partnerships to provide the environment to improve health and health services.
- **ESD to develop a Balanced Lifestyle-**
- A balanced lifestyle means being in control of all aspects of our life, such as:
- **1. Physical**
- **2. Mental**
- **3. Spiritual**
- **4. Family**
- **5. Financial**
- **6. Work and Career**
- **7. Social**

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- 8.Spiritual Practices
- 9. Sleep
- 10. Being happy-truly comes when we are in control of our own thoughts and actions.
- 11. Regular rhythm-Life and nature works in natural cycles and natural rhythms.
- 12. Simpler approach to life
- 13. Preventative Medicines- as precautionary measures
- **Role of ESD: Transformative Education**

Transformative education

For UNESCO, *transformative education* involves the teaching and learning geared to motivate and empower learners to take informed decisions and actions at the individual, community and global levels. This notably implies ensuring, for instance, the curriculum, pedagogy, learning materials, schools or learning environments are meaningful in the natural, political, economic and cultural contexts.

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Societal transformation:
Enable the achievement of the SDGs towards building a more sustainable world

Pedagogy and learning environment:
Employ interactive, project-based, learner-centred pedagogy. Transform all aspects of learning environment through a whole-institution approach to ESD to enable learners to live what they learn and learn what they live

Learning outcomes:
Empower people to take responsibility for present and future generations and actively contribute to societal transformation

Learning content:
Integrate sustainability issues, in particular those enshrined in the 17 SDGs such as climate change, into all kinds of learning

The General Assembly reaffirms education for sustainable development as an integral element of the Sustainable Development Goal on quality education and a key enabler of all the other Sustainable Development Goals.

UN General Assembly Resolution 72/222 (2017)

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- To shift to a sustainable future, we need to rethink what, where and how we learn to develop the knowledge, skills, values and attitudes that enable us all to make informed decisions and take individual and collective action on local , national and global urgencies.
- How can we prepare learners of all ages for the 'battle for our lives' as the UN Secretary-General calls it?
- ESD is a well-established framework for doing this.
- ESD empowers learners with knowledge, skills, values and attitudes to take informed decisions and make responsible actions for environmental integrity, economic viability and a just society empowering people of all genders, for present and future generations, while respecting cultural diversity.

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- ESD is recognized as a key enabler of all SDGs and achieves its purpose by transforming society
- ESD is a lifelong learning process and an integral part of quality education that enhances cognitive, social and emotional and behavioral dimensions of learning.
- It is holistic and transformational and encompasses learning content and outcomes, pedagogy and the learning environment itself.
- UNESCO recent study which reviewed policy documents of 10 countries shows that ESD is mostly associated with the teaching of scientific knowledge on environment.
- This is not enough to bring the transformative power of education to its full force

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To ensure individuals are able to understand sustainability challenges, to be aware of their relevance to the surrounding realities, and take action for change,

To trigger structural transformations in today's economic and social systems by promoting alternative values and contextualized methods,

To address the new opportunities and risks on sustainable development posed by emerging technologies,

Education needs to transform itself.

UNESCO's new global framework on ESD called *ESD for 2030* invites you to join this urgent mission.

Healthier future

Importantly, education can only be "transformative" when learners, feel valued, acknowledged, safe and are included in the learning community as full and active members. This implies paying greater attention to the quality of the learning environment and school climate, as well as to learners' political, social and cultural rights. This starts by preventing and addressing school violence and bullying, gender-based violence, as well as health and gender-related discrimination towards learners and educators. Coherence is also required between policy and practice, and also between the world we live in today at school and the world we want to build outside of school.

Education for Sustainable Development aims to raise knowledge, awareness and action in order to:

Education has to model the world, values and behaviours that we expect from children. Educators are expected to transform their teaching into something that speaks to learners, not just something that is talked about. Learners must then engage with the world and encounter it in ways that command attention and become significant.

(…) transformation necessitates, among other things, a certain level of disruption, with people opting to step outside the safety of the status quo or the "usual" way of thinking, behaving or living. It requires courage, persistence and determination, which can be present at different degrees, and which are best sourced from personal conviction, insight, or the simple feeling of what is right.

LECTURE: 24

ESD for Food Security

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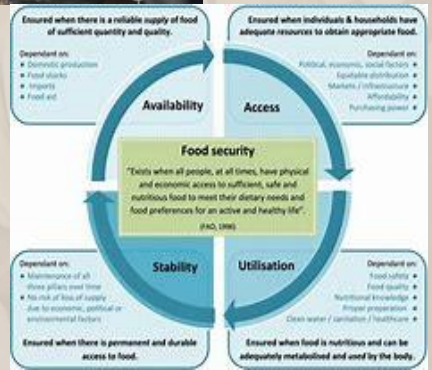
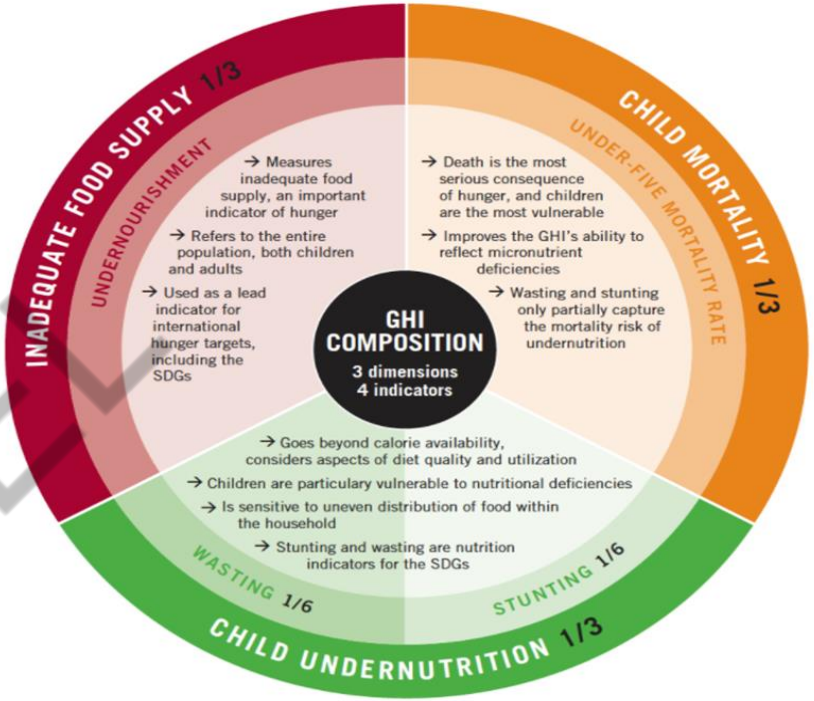
SDG 2: ZERO HUNGER



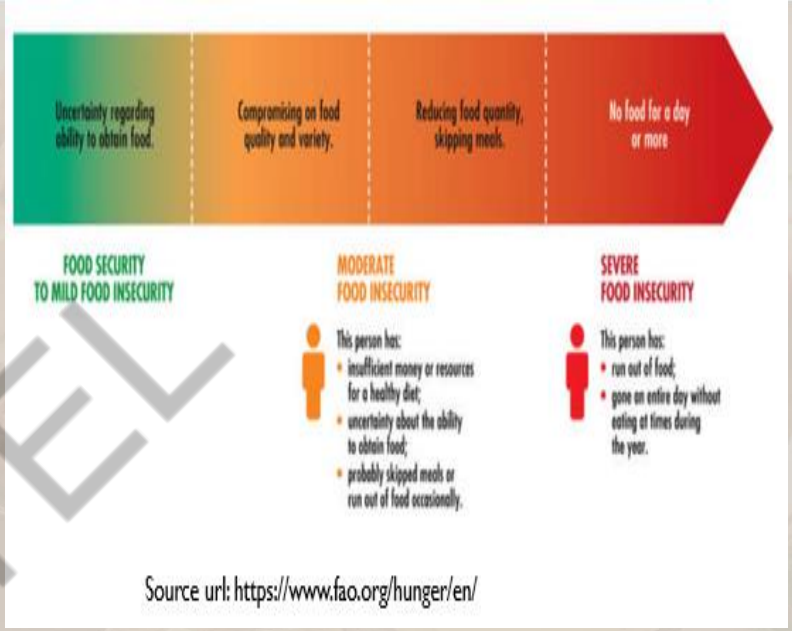
Global Hunger Index 2021

- The Global Hunger Index (GHI) is a tool designed to comprehensively measure and track hunger at global, regional, and national levels. GHI scores are calculated each year to assess progress and setbacks in combating hunger (Source: Wiesmann et al. ,2015).



Food Insecurity

- Food security exists when all people, at all times, have physical, social and economic access to sufficient, safe and nutritious food which meets their dietary needs and food preferences for an active and healthy life.
- Food insecurity occurs when a person does not have access to adequate and nutritious food on a regular basis, for healthy development and leading an active lifestyle. It is caused by unavailability of food or inadequate resources to avail food. Food insecurity is measured on the “Food Insecurity Experience Scale” (FIES).



Food Insecurity contd…

- The United Nations lists five levels of food security for any given country. Famine is the fifth and worst level. For a famine to be declared, the following three conditions must be met:
- At least 20 percent of households in an area face extreme food shortages with a limited ability to cope (fewer than 2,100 kilocalories available per day).
- At least 30 percent of the population is acutely malnourished.
- Every day, at least two out of every 10,000 people or four out of every 10,000 children die from a lack of food.

Source: istockphoto

Food Insecurity contd…

- According to the United Nations Office for the Coordination of Humanitarian Affairs (UN OCHA), Afghanistan "is currently facing the second drought in four years and the worst of its kind in 27 years".

- "An estimated 22.8 million people, or 55 per cent of the population, are expected to be in crisis or emergency levels of food insecurity (IPC 3+) between November 2021 and March 2022, a nearly 35 per cent increase from the same season last year (16.9m)".

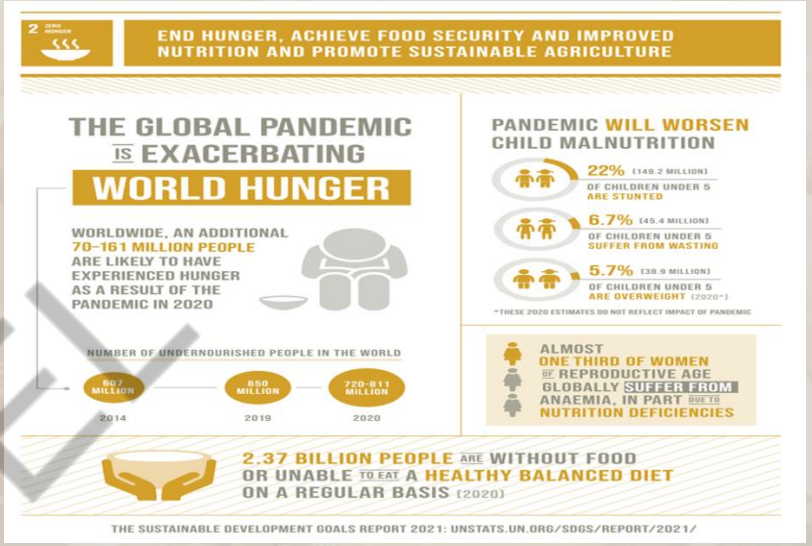
Source: HindustanTimes

Impact of COVID-19

Even before the COVID-19 pandemic hit, the world was already not on track to achieve Sustainable Development Goal 2—to eliminate hunger and all forms of malnutrition by 2030.

The immediate nutritional risk that the COVID-19 pandemic created is the increased risk of malnutrition due to the economic impact of social distancing, full or partial lockdowns, and quarantining.

Many households have suffered loss of income and/or sources of supplemental food such as school meals for children



Impact of COVID-19 Contd….

A more direct impact of COVID-19 on malnutrition has been observed. There have been increases in the incidence of malnutrition among older adults with severe COVID-19.

Pandemic-related economic contractions and disruptions to food and health systems now threaten to exacerbate maternal and child undernutrition across low- and middle-income countries (LMICs).

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Impact of COVID-19 in India

- Some COVID-19 response tactics, including physical distancing, school closures, trade restrictions, and countrywide lockdowns, impacted food systems by interrupting the production, transportation, and sale of healthy, fresh, and inexpensive foods, causing millions of households to rely on nutrient-poor alternatives.

- Healthcare systems were overburdened, and humanitarian response was disrupted. Especially when the Indian Government announced the lockdown, which resulted in the closure of Anganwadi centers and commercial organizations, it compelled the halting of healthcare outreach initiatives.

Impact of COVID-19 in India contd…..

- The mid-day meal program, which functioned as the principal source of extra nourishment for millions of school-aged children in India, was forced to halt due to school closures.
- The mass exodus of migrant workers without any food supplies made it worse. Most notably, children who already have Moderate or Severe Acute Malnutrition are now mainly living in their homes without medical attention or proper supplementary nutrition, putting their health in danger.

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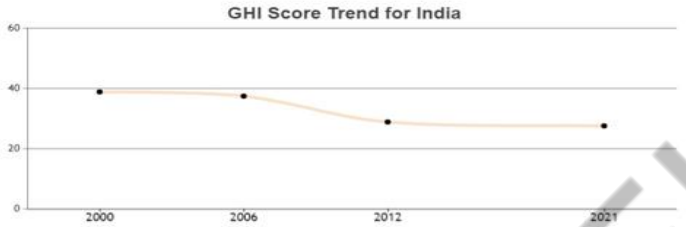
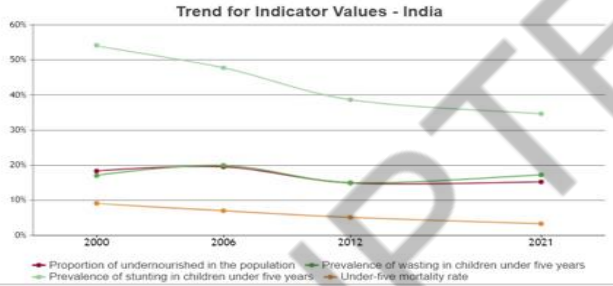
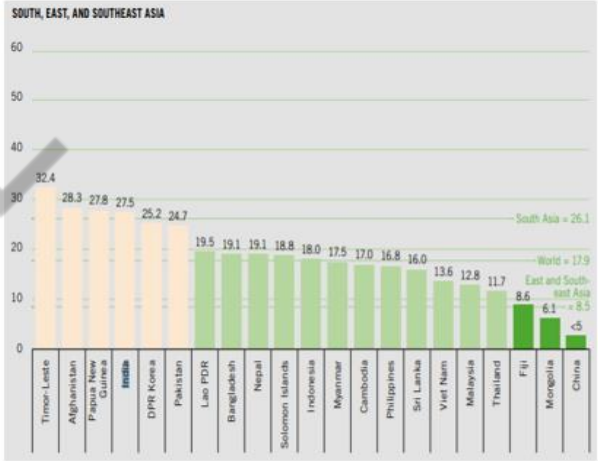
INDIA'S POSITION IN 2021







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SDG 1: Targets and Indicators

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Key Indicators of Poverty

- Landlessness
- A tiny holding with no irrigation
- Exclusive dependence of the household on agricultural employment for livelihood
- No diversification or permanent migration
- Non ownership of livestock
- Non ownership of other instruments of labour that would fetch higher wages
- Withdrawal of children from school
- Send children for work

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Key Indicators of Poverty contd…

- Very poor housing condition- unable to maintain it periodically
- Unable to source credit beyond a limit
- Large number of dependents in the household
- Belonging to SC, ST or OBC communities
- Lack of social capital to access any program meant for them sponsored by the state.
- No other skill other than agricultural labour or skills that are not in demand

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Key Indicators of Poverty contd…

- The poorest of the poor have added characteristics
- Destitute with no resource
- Unable to participate in the labour market
- Unable to seasonally migrate
- Unable to eat regularly and adequately
- Depend on others for several things including some time food.
- Children were never enrolled or withdrawn very early from schools
- Unable to access free medical help due to lack of resources

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Indicators considered for each component of Livelihoods Capital

Description of Capital
Human Capital • Labour Force Participation • Literacy and Education • Health
Natural Capital • Land Holding • Livestock Holding • Common Property Resources (CPRs)
Physical Capital • Instruments of Labour • Consumer Durables • Irrigation Instruments • Community owned Physical Capital
Financial Capital • Savings • Debt
Social Capital • Self Helping Groups(SHGs) • Grama Saba

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LECTURE: 25

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Vulnerability of the Poor

Decline in Agricultural Employment
vulnerability of the Poor
Decline of Dry Land Agriculture
Competing Demand for Water
Lack of Water and Marginal Holdings
Impact of Reduced Employment and Livelihoods of the Poor
Coping Mechanism : Seasonal unemployment
Adaptation as a Livelihood Strategy
Health induced Vulnerability
Impact of Ill Health
Life Cycle Rituals :Marriages, deaths and -
other such major events in the family

Vulnerability of the Poor contd…

The major problem that poor encounter in their everyday life is the security of their livelihood.

Stresses and shocks abound their livelihoods resulting in serious undermining of their very sustenance.

There are certain shocks and stresses that they can manage on their own but there are very many others that they are ill equipped to negotiate.

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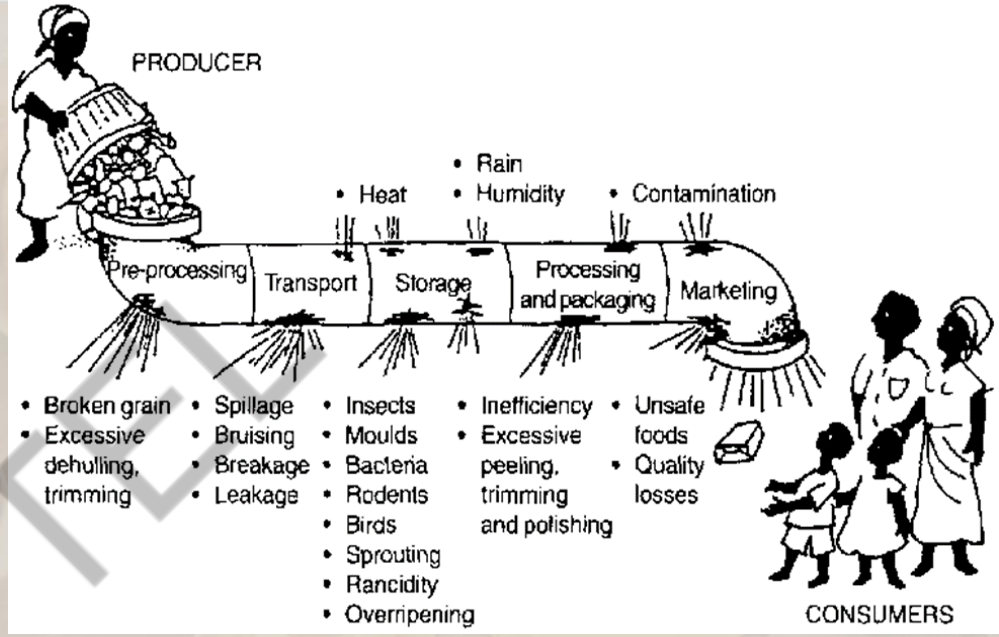
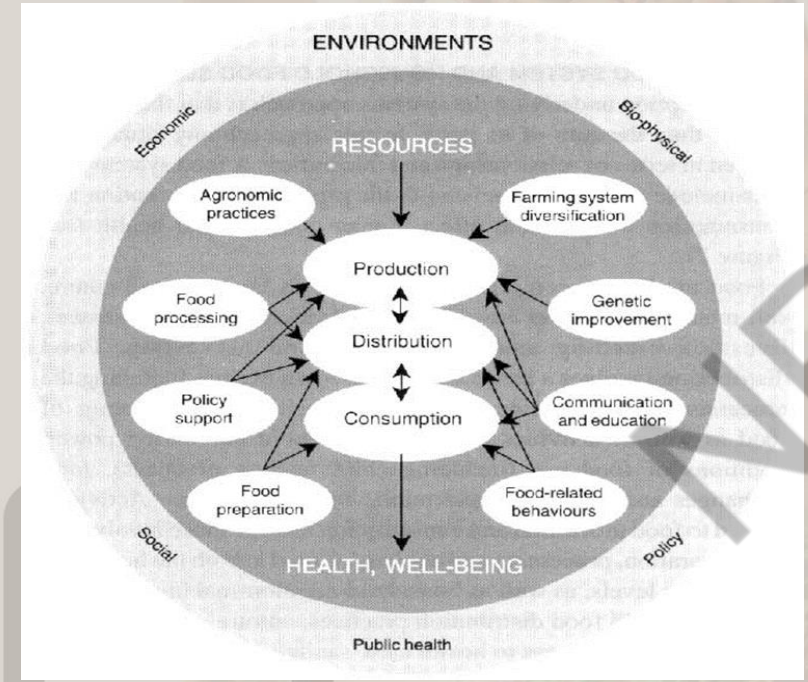
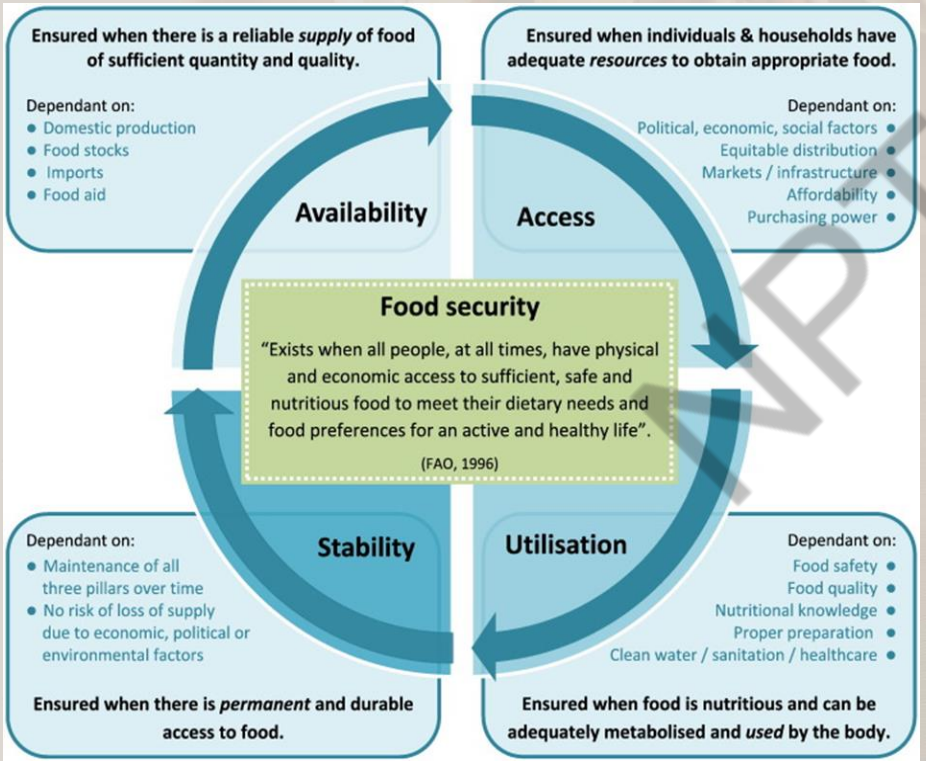


The food pipeline
Source: Adapted from Pariser, 1987



The food system
Source: Combs et al., 1996.

Three Pillars of Food Security



- The three pillars of food security as defined by the World Health Organization (WHO) to be:
- 1. Food Access – Having sufficient resources to obtain appropriate foods for a nutritious diet.
- 2. Food Availability - Sufficient quantities of food available on a consistent basis.
- 3. Food use - Appropriate use based on knowledge of basic nutrition and care, as well as adequate water and sanitation.
- These pillars were implemented after the UN stated rules about the minimum dietary requirements of an individual to live a substantial life.
- Figure shows the pillars of food security and then the aspects of those that compromise them. These are sub-pillars that are all interlinked with each other.

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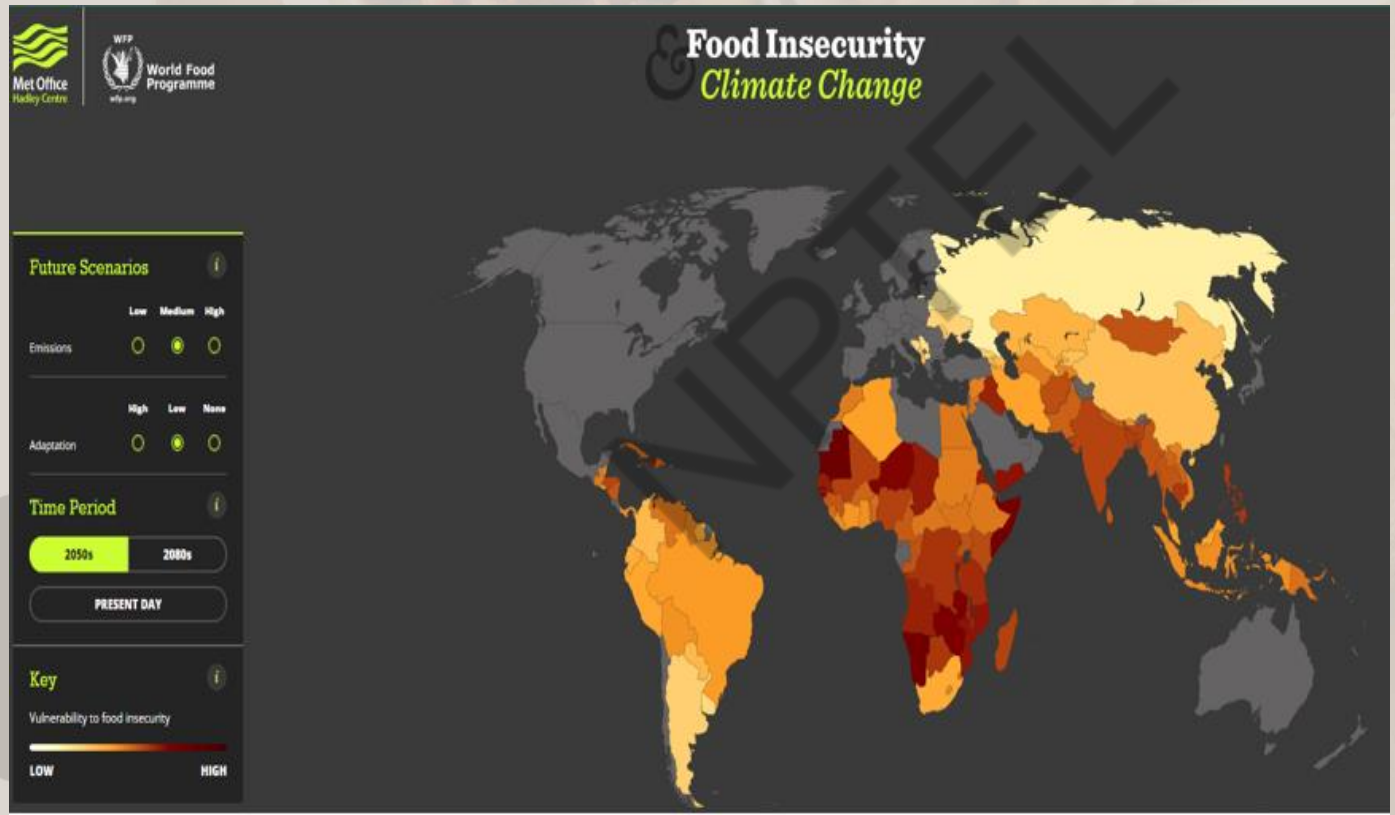
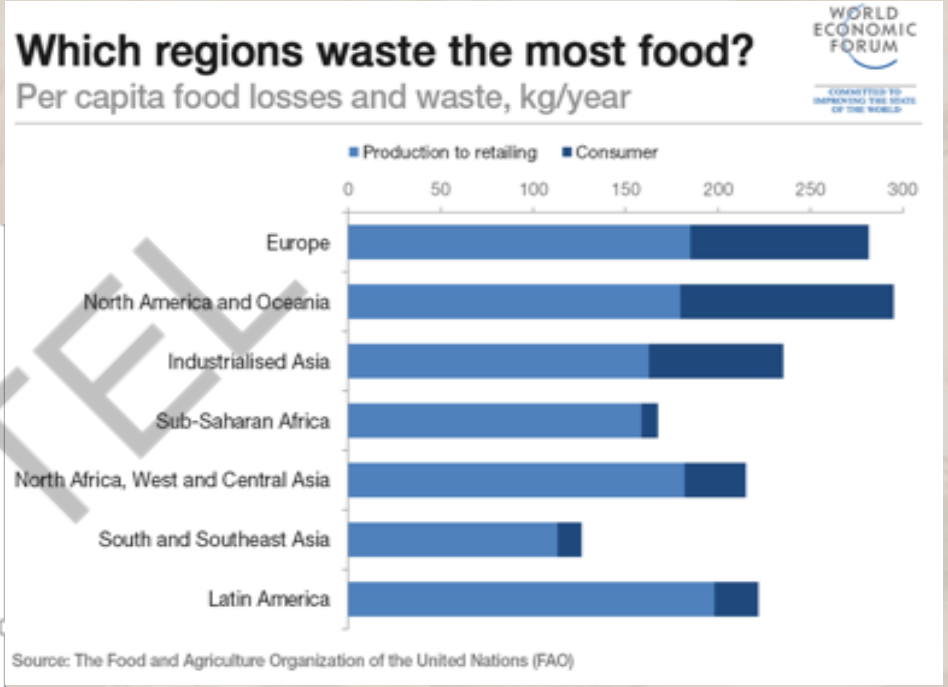

Met Office
Hadley Centre
WFP
World Food Programme
wfp.org
Food Insecurity
& Climate Change
Future Scenarios
Low
Medium
High
Emissions
High
Low
None
Adaptation
Time Period
2050s
2080s
PRESENT DAY
Key
Vulnerability to food insecurity
LOW
HIGH

Food Shortages and Waste

- Food shortages in developing countries are common. The people most affected are smallholder farmers and their families who depend on their own surplus to survive between harvests.
- The period leading up to a harvest is known as the "hungry season." Food from the previous harvest runs out and families cut back on meals. This period of time may last for months depending on the size of the previous harvest.
- Around the world families with very low incomes run out of money at the end of the month. Families cut back on how much they eat and then eventually skip meals altogether on some days.
- Another reason for food shortages is up to 40 percent of food grown in some countries is spoilage. Smallholder farmers do not have adequate storage facilities to protect their supplies against pests and weather.



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